

Classification of Nutrient Deficiencies in Plants Using Recurrent Neural Network

Ramasamy S
Associate Professor
Department of CSE
Malla Reddy College of
Engineering and Technology
Hyderabad, Telungana.
ramasamy_s@gmail.com

Chandrasekar V
Professor
Department of CSE
Malla Reddy College of
Engineering and Technology
Hyderabad, Telungana.
drchandru86@gmail.com

Viswa Bharathy A M
Assistant Professor
Department of CSE
GITAM University
Bengaluru, Karnataka.
viswabharathy86@gmail.com

Abstract – The symptoms associated with deficiencies in plants tends to appear often on the leaves. The color and shape of a leaf often used for diagnosing the nutritional deficiencies in plants and classification of these properties often pose serious problem. Since same color and shape may have many root cause problems. It is hence necessary to carefully analyze the texture of leaf with proper training of a classifier. In this paper, we design an acquisition-based classification model that utilizes Internet of Things (IoTs) for data acquisition and recurrent neural networks (RNN) for the task of classification. Prior classification, the model is trained over several iteration based on careful observation of features and its related symptoms. The simulation is conducted with fine-tuning of classification after several iterations. The results of simulation show that the proposed method obtains improved classification accuracy in terms of accuracy and F-measure than other deep learning models.

Index Terms - Nutrient Deficiencies, Plant Leaf, Recurrent Neural Network, Classification

I. INTRODUCTION

Fertilizers are considered as a critical commodity to worldwide production of food crops, especially in guaranteeing high and consistent high yield of rice [1]. When appropriate fertilizers are applied in the right quantity at the right time, the best results are achieved. In Asian countries, rice is frequently grown crop without the appropriate nutrient input. Non-traditional procedures are ubiquitous, and if it is combined with a gap between inventions and widespread technological adoption, rice crop receives an unbalanced nutrient application. Blind fertilization is still a common practice today. As a result, with increasing fertilizer quantity, the generation of rice yield gains tends to get reduced and hence the rice quality degrades [2].

Since the soils nutrition fail to meet the demands of nutrient in growing rice crops, and hence

diagnosing the nutrient deficits in rice is an important element of scientific fertilization. The identification of required nutrients will aid in the creation of a fertilization program that will provide the target nutrients while avoiding overstock of others.

Rice nutrient deficiency symptoms are signs of rice crop malnutrition and based on the morphology of rice plants it can be visually assessed. Further it provides enough data for the detection of nutrient deficiency [3] [4]. On other hand, if the agricultural fields are large, the problem further tends to get increased and many nutrient inadequacies are spread out over a farm or region, as well as over time during the growing season. As a result, agricultural experts are finding it impossible to meet the service demand, preventing nutritional deficiencies from being adequately recognized and efficiently addressed.

Nondestructive analytical and chemical methods are employed to diagnose nutrient deficits in crops in addition to manual diagnosis. Full-scale analysis or fast tissue measurement are the two types of chemical diagnosis [5]. A full-scale examination determines how much of each nutrient element is present in a crop. This diagnostic method may identify all of the nutrients required for a plant survival as and crop growth [6]. The yield prediction is consistent and accurate, and they provide enough information for a nutrient diagnosis, however the approach is labor intensive and limited to a laboratory.

Visual colorimetric is a simple and a quick technique suited well for field diagnosis, is used to quickly measure unassimilated nutrients in plant tissues [7]. This method is typically assess the shortages of important elements such as nitrogen, phosphorus and potassium ions in crops because to the considerable investigation necessary for deeper analysis. The low content of trace elements, as well as the high precision necessary for their examination, makes it difficult to determine their status quickly.

In site measuring with a spectral remote sensing, chlorophyll metre and computer vision are examples of nondestructive diagnosis procedures. The first two methods are time-consuming or difficult to implement in terms of data processing, and they are only appropriate for certain aspects [8, 9]. For low-cost image collecting and transport, computer vision, on the other hand, requires only an IoTs, and enabling image diagnostics perhaps the most versatile alternatives.

The image recognition capabilities of CRNNs are impressive and the versatility of open source development frameworks substantially aids the evolution from plans and concepts to applications and outcomes, lowering the obstacles to application development. Prior research may tend to resolve the technical difficulties like decay in learning rate or hyper parameter or scheduling of crop yield. Therefore, new vision-problem applications can commonly influence the topologies to aid the yield of crop development.

Due to the substantial time investment, optimized settings of hyper parameter are considered as a significant task. As the number of apps generated and shared grows, automatic model design is a viable option. Although some CRNNs may rescale image sizes, the spatial resolution conditions of the rice leaf images must be considered. The recognition accuracy will be harmed if the recognized object region in the overall image is too small. Furthermore, the backdrop image might have an impact on recognition accuracy, particularly in multiclass identification. Unlike picture recognition tasks that entail identifying people or vehicle items, identifying nutrient deficits in rice leaves necessitates recognizing small textural changes in often similar photos. Because the goal of this study is recognizing nutritional deficits in rice from multiple images, surface texture should be considered as an identification feature.

In this paper, IoTs sensors are used to collect the images of rice plant via data acquisition module. The images are then pre-processed and the features are extracted for optimal classification by the convolution recurrent neural networks.

II. RELATED WORKS

Many studies have looked into using them to diagnose agricultural diseases. Mohanty et al. [10] used images of plant leaves to train a DCNN with deep learning to identify crop illnesses in various species. The study used two designs, Alex Net and

Google Net, to assess the viability of DCNNs for crop disease diagnosis. Both models obtained up to 99.4% accuracy. The trained model accuracy dropped significantly to 31.4% when other labeled images taken under other settings were used, although it was still significantly higher than random selection. As a result, the deep learning is extremely accurate but it failed in its robust. This could explain why developing a robust model is difficult. The dataset should be as large as possible because observable symptoms are time-specific schemes.

Brahimi et al. [11] used transfer learning to train Google Net and AlexNet on own dataset. They were able to identify nine kinds of tomato illnesses with 99% accuracy. SVM and random forest were shown to be less accurate than their techniques. Liu et al. [12] correctly identified four disease classes on apple leaves with increased accuracy, and Lu et al. [13] correctly classified rice into 10 classes of disease with 95.5% accuracy. Zhang et al. [14] used upgraded GoogLeNet to detect nine kinds of maize leaf diseases with a 98.9% accuracy.

All of these investigations obtained high crop disease classification accuracy, highlighting the utility of DCNNs. It difficult to create a really complete database when it comes to data. The appearance of symptoms can change as crop illnesses or nutrient deficits progress, and they can also be affected by environmental conditions like humidity and temperature. To cover the complete spectrum of options, a huge number of images are required. Another factor to consider is that all photos must be appropriately labeled, which is frequently time-consuming [15]. Uneven sampling can lead to the poor optimization but weighted cross-entropy loss [16] [17] has been presented as a solution to this problem. In terms of features, substitution settings and feature reuse can further reduce the number of parameters while it partially reduces the gradient disappearance problem.

For identifying the nutrient deficits in case of black gram leaves, Watchareeruetai et al. [18] achieved the precision rate of 43%. It challenging to categories distinct forms of nutrient shortages because the dataset contains varied deficient periods of plant leaves.

Sethy et al. [19] developed a pre trained DCNNs model that helps in accurately identifying the deficient in plants along with a boosting strategy using an SVM. This offers an accuracy rate of 99.8% accuracy rate. This motivates us to investigate the

capacity of DCNNs to classify additional components in rice with distinct deficiency phases.

III. PROPOSED METHOD

In this section, Figure 1 show the illustration of the proposed method, where the details of which are given below:

Collection

IoT sensors employ a camera to collect the intricacies of plant leaves and examine those using benchmarked datasets. The nutrient-deficient rice leaves used in this investigation were obtained from a variety of government agencies and websites. Different augmentation approaches are employed to make the dataset large enough to develop the model.

Image Pre-processing

Any computer vision-based model requires image pre-processing to train, validate, and test it. It will remove any extraneous data or distortions while enhancing the key elements required to building the model. The acquired photos will go through an image preparation stage, which will improve the image analysis accuracy.

Resizing the image

The size of the input image should match the size of the database images. A major necessity for building a computer vision-based model is to create a base size for all images. For this investigation, the image dimensions are between 560x480 pixels.

Enhancing the image

In this phase, the contrast of the images is adjusted so that the items stand out more from the backdrop and other objects. In low contrast images, histogram equalization is used to increase contrast. It is used to enhance the image contrast, color, and brightness.

The noise in the original image will have undesirable outcomes. Different picture filters can be applied to reduce noise and enhance the image edges. We are utilizing a bilateral filtering strategy in this study to retain edges. The intensity of each pixel value is replaced by the weighted average of its neighbors in this smoothing approach.

Feature Extraction

The algorithm is given a large amount of data, which can lead to complicated processing. The

specified inputs are compacted and fused together to form a set of features. If the image characteristics are extracted correctly, it can extract the necessary information from the input to complete the task at hand. The most important aspect of texture analysis is feature extraction. The leaves' statistical attributes are extracted using a GLCM-based colour co-occurrence approach.

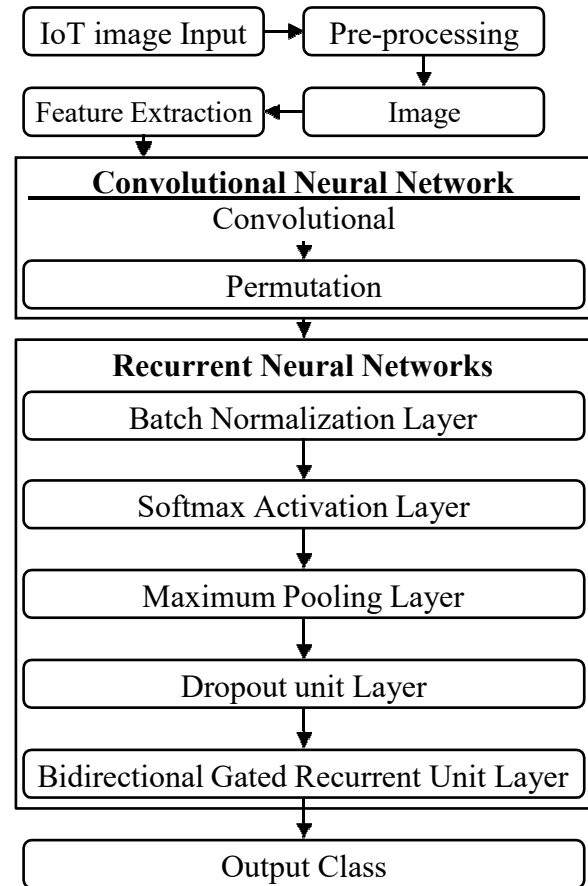


Figure 1: CRNN Framework for classification

The selected features are Variance, Contrast, Energy, Correlation, Homogeneity, Smoothness, Entropy, Mean, RMS contrast Standard Deviation, skewness, and Kurtosis. All these are calculated from GLCM using its corresponding formulas.

Classification

During the testing phase of our classifier engine, classes of the rice crops are retrieved. The CNN classifier receives the residual data with plant leaf as an input.

The convolution layer helps in the extraction of the plant leaf from the input image using Image Net training based on the extracted features.

1. When a weight sequence is multiplied with input data array, a convolution layer tends to proceed with the processing operation which is of a linearone.
2. To detect the nutrition deficiency in the input leaf image, adoption of filter is conducted that helps in scanning the entire image and this is referred as the process of translation invariance. A two-dimensional feature map is formed because filters are applied repeatedly to the input array.
3. The convolution layer lists the total number of labeled items from the feature space in order of providing permutation-based representation and that helps in classifying the input image using this permutation layer.
4. Instead of utilizing the distance function, the similarity is calculated using the two permutations. The ground truth value is considered between the ground truth and reference object. This relationship is the most similar to the one revealed by distance.

The operations of RNN for checking the accuracy of detected classes is given below:

1. Batch normalization cuts down certain hidden layers needed to process CNN input information. To increase the process of learning, it modifies the scaling and activation functions.
2. Batch normalization reduces certain hidden layers in CNN for the purpose of input processing. It further adjusts the scaling and activation functions to accelerate the learning process.
3. Soft max Activation is considered as a generation of a vector function with possible availability of probabilistic outcomes.
4. Maximum pooling is a procedure that helps in determination the greatest or maximum value at each feature map in each patch, and it further helps in the forward pass and permutation layers to reveal the information.

5. The Bidirectional Gated Recurrent Unit is utilized to alleviate the gradient problem and make operations quick and efficient.
6. On the forward pass, the dropout unit ignores a number of hidden units that helps in reducing the individual nodes, resulting in a reducing the hidden nodes. The start and output edges of a dropped hidden node are hence considered removable, where it helps to overcome the process of over fitting.

IV. RESULTS AND DISCUSSIONS

In this section, we conduct the experiments on a high-end computing system with intel Core i9 7920X CPU that runs on a 64 GB RAM with two different GPUs namely GeForce GTX 1080Ti of 11 GB memory. Keras framework is used for implementing the module with the Tensor Flow backend. To attain model fitting, the CRNN model is fine-tuned with the pre trained weights using Image Net without data augmentation. The model evaluation is conducted using kappa score and accuracy metric.

A three-fold cross validation is conducted to find the average accuracy of the CRNN during training and testing phases. Further, the training process is accelerated using an Adam Optimiser and Python Image Library is used for image resizing in order to fit the model requirement.

The proposed CRNN is compared with DCNN, ResNet and Dense Net. Here the proposed model involves reading the color features from the images collected via IoT devices and then training with CRNN.

Experiments comparing several learning rates revealed that 0.00001 was the best learning rate for the model to run 50 epochs with no decay. Batch size should not exceed 10 due to certain constraints associated with GPU on CRNN. As a result, we used the same batch size and learning rate for all of the tests to evaluate the models in terms of time efficiency. The study is further tested with three different indices that estimate the prediction accuracy of CRNN and it includes the following:

The recall rate is defined as the ratio of predicted positive cases with entire cases in actual class. The results of simulation are shown in Figure 2, where the proposed method achieves higher rate of recall than the conventional methods.

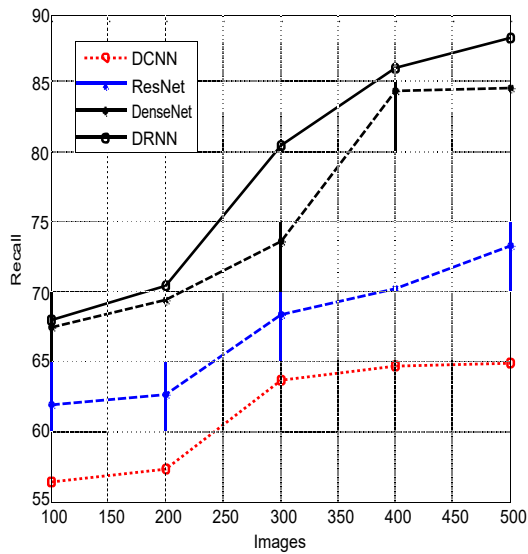


Figure 2: Recall

The precision rate is defined as the ratio of predicted positive cases the total predicted cases. The results of precision rate are shown in Figure 3, where the proposed method achieves higher rate of precision than the conventional methods.

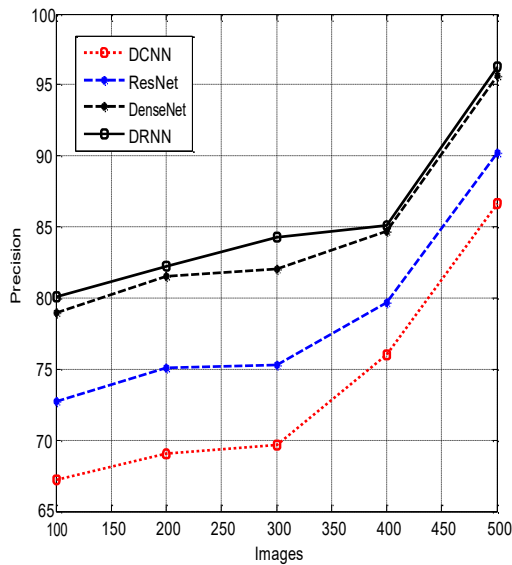


Figure 3: Precision

The f1-measure is defined as the weighted average of both precision and recall indices to identify the accuracy of the model with uneven class distribution.

The results of f1-measure are shown in Figure 4, where the proposed method achieves higher rate of f1-measure than the conventional methods.

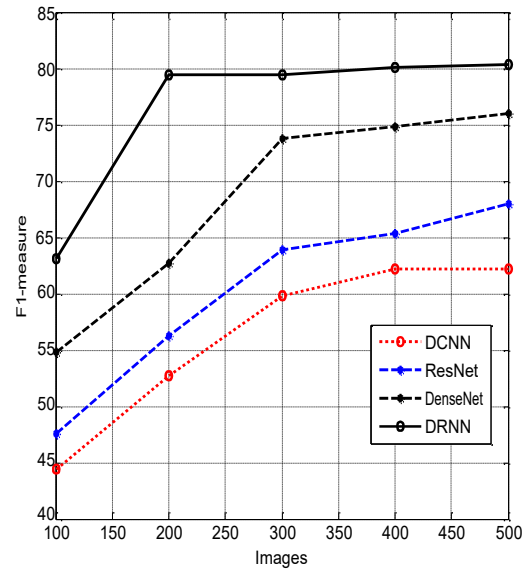


Figure 4: F1-measure

The CRNN is trained in such a way that it identifies the nutrient deficiencies in terms of various classes. The model is made effectively fitted after the process of tuning and the accuracy of all the training, testing and validation is higher than 90% with kappa scores higher than 0.90. The results of simulation show that the CRNN model achieves an average accuracy higher than the existing models with increased precision, recall and f-measure rate.

In terms of the proposed approach, CRNN test and validation accuracy increased slowly, which could be attributed to the large number of parameters in this model compared to other models. As a result, the pre trained parameters are progressively updated during back propagation to achieve strong generalization ability. Furthermore, the CRNN was able to adequately fit the data, but traditional approaches appeared to over fit the data.

The explanation for this could be that some of the labeled photos featured complicated backgrounds, and the outline data in the ResNet and Dense Net approaches would easily leads to over fitting. In higher-level networks, the CRNN technique, on the other hand, can synthesize local receptive fields.

The prediction results of various nutrient deficiencies are shown in Table 1, where the prediction of different macronutrients and micronutrients are given in terms of f1-scores, where the result of Fe is found to be worst.

Table 1: Prediction accuracy of various macro or micronutrients

Class	Recall	Precision	f1	Samples
Potassium (K)	0.98	1.00	0.99	92
Phosphorus (P)	1.00	0.97	0.98	61
Nitrogen (N)	0.98	1.00	0.99	43
Sulfur (S)	0.95	0.95	0.95	42
Calcium (Ca)	1.00	1.00	1.00	43
Magnesium (Mg)	1.00	1.00	1.00	78
Cuprum (Cu)	0.98	0.96	0.97	104
Ferrum (Fe)	0.83	0.94	0.88	31
Zinc (Zn)	0.97	0.96	0.97	112
Manganese (Mn)	0.97	1.00	0.98	64
Full	0.96	0.90	0.93	58

Deficiency is a relative concept, and a slight deficiency might be mistaken for full nutrition. In this study, another area for misclassification was among Fe, Mn, and Zn deficiencies which all shared some common symptoms.

These nutrients are directly or indirectly related to chlorophyll formation or photosynthesis, the disruption of which generally causes chlorosis. Furthermore, Zn deficiency was misclassified as Si deficiency as both of them caused brown spots in rice leaves. These noticeable values can be extracted from the confusion matrix (Figure 5) over the datasets in Table 2 under different classes:

Class 0: Healthy
Class 1: Chlorosis
Class 2: Brown Spots
Class 3: Septoria Spot

Table.2. Dataset Size

Class Label	Class	Data Set Size	Training Set Size	Test Set Size
0	Healthy	483	387	96
1	Chlorosis	456	360	96
2	Brown Spots	431	350	81
3	Septoria Spot	447	356	91

True Label	Predicted Label				
	Class	0	1	2	3
	0	92	0	4	0
	1	2	83	8	3
	2	0	8	70	3
	3	3	1	9	78

(a) DCNN

True Label	Predicted Label				
	Class	0	1	2	3
	0	92	2	2	0
	1	2	85	7	2
	2	1	4	71	5
	3	2	3	7	79

(b) ResNet

True Label	Predicted Label				
	Class	0	1	2	3
	0	93	1	2	0
	1	1	86	6	3
	2	0	1	74	6
	3	1	0	5	85

(c) Dense Net

True Label	Predicted Label				
	Class	0	1	2	3
	0	93	0	3	0
	1	2	89	3	2
	2	0	2	73	6
	3	1	0	6	84

(d) CRNN

Figure 5: Confusion Matrix

V. CONCLUSION

In this paper, IoTs sensors are used to collect the images of rice plant via data acquisition module. The images are then pre-processed and the features are extracted for optimal classification by the convolutional recurrent neural networks. The training is conducted using the test dataset on various classifiers and then the trained model is tested with 20% of the collected data. The trained model helps in classification of instances i.e. classes of cooked rice types (texture) that includes.

The results of simulation show that the proposed model obtained improved class of accuracy than other methods. Further, the classification errors during training and testing are low in proposed model than the existing methods. The results thus show a promising model for an automated nutrient deficiency classification in case of rice plants.

REFERENCES

- [1] Xu, X., He, P., Yang, F., Ma, J., Pampolino, M. F., Johnston, A. M., & Zhou, W., "Methodology of fertilizer recommendation based on yield response and agronomic efficiency for rice in China". *Field crops research*, 206, 33-42, 2017.
- [2] Liu, X., Zhang, Y., Han, W., Tang, A., Shen, J., Cui, Z., ... & Zhang, F., "Enhanced nitrogen deposition over China", *Nature*, 494(7438), 459-462, 2013.
- [3] Malav Jugal, K., & Ramani, V., "Effect of silicon on nitrogen use efficiency, yield and nitrogen and silicon contents in rice under loamy sand soil". *Res J Chem Environ*, 21(4) . 2017.
- [4] Zhang, Y., Wang, Y., & Niu, H. (2017). Spatio- temporal variations in the areas suitable for the cultivation of rice and maize in China under future climate scenarios. *Science of the Total Environment*, 601, 518-531.
- [5] Ata-Ul-Karim, S. T., Cao, Q., Zhu, Y., Tang, L., Rehmani, M. I. A., & Cao, W., "Non- destructive assessment of plant nitrogen parameters using leaf chlorophyll measurements in rice", *Frontiers in Plant Science*, 7, 1829, 2016.
- [6] Xiong, X., Zhang, J., Guo, D., Chang, L., & Huang, D., "Non-Invasive Sensing of Nitrogen in Plant Using Digital Images and Machine Learning for Brassica Campestris ssp", *Chinensis L. Sensors*, 19(11), 2448, 2019.
- [7] Anami, B. S., Malvade, N. N., & Palaiah, S., "Classification of yield affecting biotic and abiotic paddy crop stresses using field images", *Information Processing in Agriculture*, 7(2), 272-285, 2020.
- [8] Etesami, H., & Jeong, B. R., "Importance of silicon in fruit nutrition: agronomic and physiological implications", In *Fruit Crops* (pp. 255-277), 2020.
- [9] Balasubramanian, V., Morales, A. C., Cruz, R. T., Thiagarajan, T. M., Nagarajan, R., Babu, M., ... & Hai, L. H., "Adaptation of the chlorophyll meter (SPAD) technology for real-time N management in rice: a review", *International Rice Research Notes*, 25(1), 4-8, 2000.
- [10] Mohanty, S. P., Hughes, D. P., & Salathé, M., "Using deep learning for image-based plant disease detection", *Frontiers in plant science*, 7, 1419, 2016.
- [11] Brahimi, M., Boukhalfa, K., & Moussaoui, "A Deep learning for tomato diseases: classification and symptoms visualization". *Applied Artificial Intelligence*, 31(4), 299-315, 2017.
- [12] Liu, B., Zhang, Y., He, D., & Li, Y., "Identification of apple leaf diseases based on deep convolutional neural networks". *Symmetry*, 10(1), 11, 2018.
- [13] Lu, Y., Yi, S., Zeng, N., Liu, Y., & Zhang, Y., "Identification of rice diseases using deep convolutional neural networks", *Neurocomputing*, 267, 378-384, 2017.
- [14] Zhang, X., Qiao, Y., Meng, F., Fan, C., & Zhang, M., "Identification of maize leaf diseases using improved deep convolutional neural networks". *IEEE Access*, 6, 30370-30377, 2018.
- [15] Barbedo, J. G., "Factors influencing the use of deep learning for plant disease recognition", *Bio systems engineering*, 172, 84-91, 2018.
- [16] Aurelio, Y. S., de Almeida, G. M., de Castro, C. L., & Braga, A. P., "Learning from imbalanced data sets with weighted cross-entropy function", *Neural processing letters*, 50(2), 1937-1949, 2019.
- [17] Cai, Z., Fan, Q., Feris, R. S., & Vasconcelos, N., "A unified multi-scale deep convolutional neural network for fast object detection", In *European conference on computer vision* (pp. 354-370), 2016.
- [18] Watchareerutai, U., Noinongyao, P., Wattanapaiboonsuk, C., Khantiviriya, P., & Duangsrirai, S., "Identification of plant nutrient deficiencies using convolutional neural networks", *International Electrical Engineering Congress (iEECON)* (pp. 1-4). IEEE, 2018.
- [19] Sethy, P. K., Barpanda, N. K., Rath, A. K., & Behera, S. K., "Nitrogen deficiency prediction of rice crop based on convolutional neural network". *Journal of Ambient Intelligence and Humanized Computing*, 11(11), 5703-5711, 2020.